CELON PHARMA

Fox Evolution: A Demonstration Report

Simulated morphometric analysis of three fox species

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Demonstration document

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1 Summary

Three fox species (red, Arctic and fennec fox) were compared on simulated morphometric data from 150 specimens collected at 6 sites and spanning the last 3.5 million years (Ma). Body mass changed by -3.4% per million years of specimen age in the red fox, and ear length increased by 0.13 cm per °C of mean annual temperature (Section 4). Lineage accumulation differed between three fictional clades (Figure 5). The description is exploratory and purely illustrative.

2 Introduction

2.1 Background

Foxes are small to medium-sized members of the family Canidae and occupy habitats from Arctic tundra to hot desert [1]. Their appendages are a textbook illustration of adaptation to ambient temperature: large ears in the fennec fox dissipate heat, whereas the compact ears of the Arctic fox conserve it [2]. Figure 1 shows a historical painting of common foxes in a winter landscape.



Figure 1: Common foxes in the snow. Painting by Wilhelm Kuhnert (public domain, Wikimedia Commons).

2.2 Objectives

The objectives of this demonstration analysis were to

1. describe the change of body mass over geological time,
2. relate ear length to mean annual temperature, and
3. compare the accumulation of lineages between clades.

3 Methods

3.1 Specimens and data sources

Specimen records were simulated for three species (Table 1). Each record holds the site, the specimen age, the mean annual temperature at the site and time of deposition, and five morphometric measurements[1]. Sampling bias is a known problem of fossil data [3] and is ignored here.

Table 1: Simulated specimens by species: number of specimens, age range, and mean (SD) of body mass and ear length.

Common name	Species	Habitat	N	Age range (Ma)	Body mass (kg)	Ear length (cm)
Red fox	*Vulpes vulpes*	Temperate woodland	60	0.1–3.5	6.23 (0.53)	9.9 (0.7)
Arctic fox	*Vulpes lagopus*	Arctic tundra	45	0.1–3.5	3.36 (0.30)	7.1 (0.8)
Fennec fox	*Vulpes zerda*	Hot desert	45	0.2–3.3	1.12 (0.09)	15.5 (0.7)

Values are arithmetic mean (standard deviation).

3.2 Statistical analysis

Body mass m of specimen j of species s was modeled on the log scale as a linear function of specimen age t,

$$\log m_{sj} = \beta_{0,s} + \beta_{1,s}\, t_{sj} + \varepsilon_{sj}, \qquad \varepsilon_{sj} \sim N(0, \sigma^2), \tag{1}$$

and ear length was modeled analogously on mean annual temperature. Both models contain a species-specific intercept and slope (details in Section A1). Lineages through time were simulated as a pure-birth process [4]. Skull shape scores were derived as described by [5].

> **Note**
>
> All p-values are descriptive. No hypothesis was specified in advance, so no multiplicity adjustment was made.

[1] Body mass, ear length, skull length, tail length and hind foot length.

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4 Results

4.1 Body mass through time

Body mass was highest in the red fox and lowest in the fennec fox, with a slight increase towards the present in all three species (Figure 2). On the log scale the red fox slope was -0.035 per million years (Table A1-1).

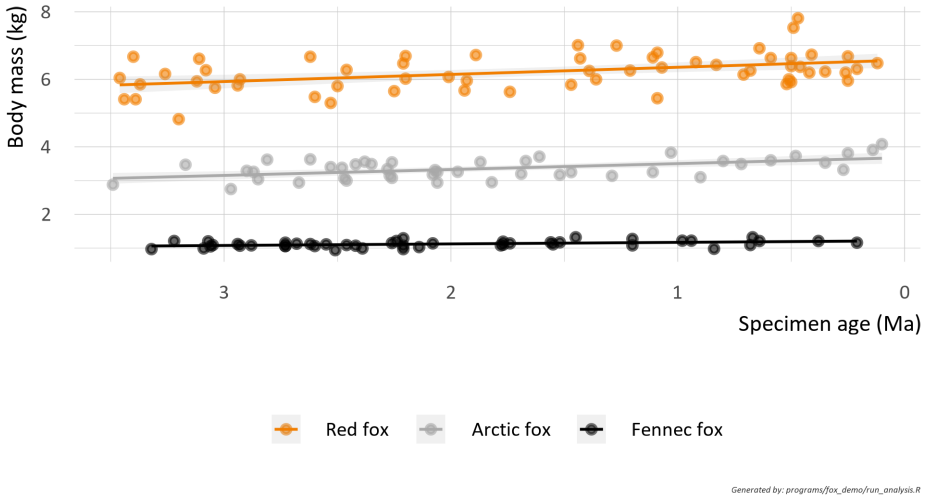


Figure 2: Body mass by specimen age and species with linear fit and 95% confidence band.

4.2 Ear size and climate

Ear length increased with mean annual temperature in every species (Figure 3), in line with the heat-dissipation argument of [2]. The fennec fox (Figure 4) had the longest ears at every temperature.

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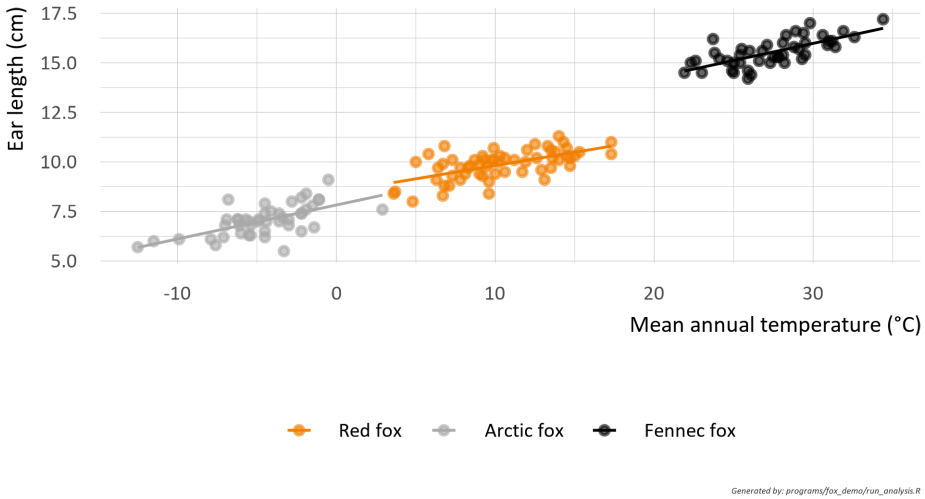


Figure 3: Ear length by mean annual temperature and species with linear fit.



Figure 4: Fennec fox (*Vulpes zerda*). Photograph by Drew Avery, CC BY 2.0, Wikimedia Commons.

4.3 Diversification

The three fictional clades accumulated lineages at different rates (Figure 5). Clade I, the fastest, accumulated the most lineages, whereas Clade III grew only modestly.

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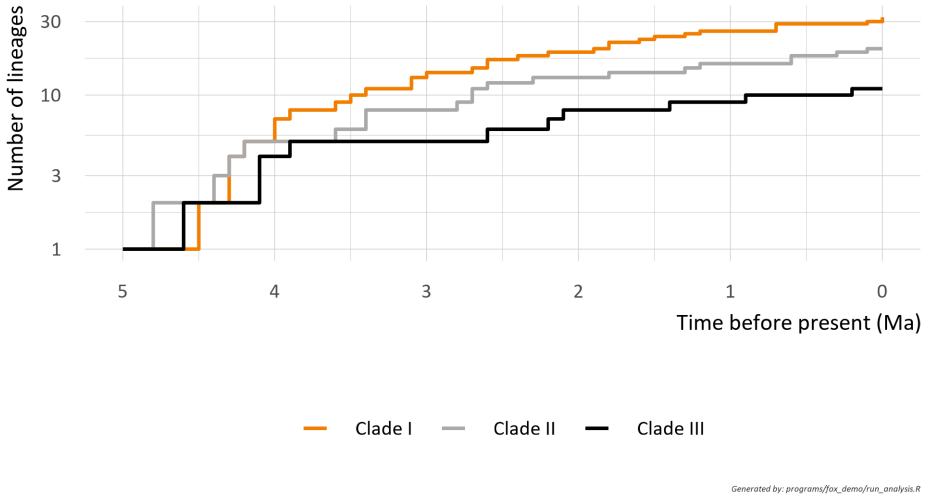


Figure 5: Lineages through time (log scale) for three fictional clades.

4.4 Specimen-level data

Table 2 lists a selection of specimens with all measurements. The table occupies a full landscape page; the page orientation returns to portrait after it.

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Table 2: Selection of simulated specimens (eight per species) with provenance, context and morphometric data.

	Provenance			Context		Morphometrics					Shape scores	
Specimen	Site	Region	Latitude (°N)	Age (Ma)	Temp. (°C)	Body mass (kg)	Ear (cm)	Skull (cm)	Tail (cm)	Hind foot (cm)	Score 1	Score 2
Red fox												
FX-001	Site E	Arabia	25	2.20	14.7	6.69	9.8	13.0	55.8	16.8	1.43	0.93
FX-002	Site A	Northern Europe	58	1.11	12.5	6.65	10.9	12.8	53.4	16.1	1.45	0.52
FX-003	Site A	Northern Europe	58	0.92	9.9	6.51	10.7	13.1	54.0	15.7	0.78	0.57
FX-004	Site F	Alaska	65	2.93	14.0	6.00	10.1	11.8	57.1	15.7	0.72	1.71
FX-005	Site E	Arabia	25	0.50	9.2	6.63	10.3	13.3	54.6	16.2	1.31	0.30
FX-006	Site C	Greenland	72	0.12	5.0	6.48	10.0	12.7	56.1	16.4	1.20	0.92
FX-007	Site D	Sahara	24	1.44	6.7	7.01	9.9	12.9	56.4	17.1	0.88	0.46
FX-008	Site D	Sahara	24	1.39	9.2	6.25	9.3	12.7	53.9	16.8	1.33	0.62
Arctic fox												
FX-061	Site D	Sahara	24	2.87	-1.4	3.26	6.7	10.7	45.6	13.1	-0.71	0.53
FX-062	Site A	Northern Europe	58	2.07	-4.5	3.31	6.5	10.2	46.4	13.9	-1.51	-0.13
FX-063	Site F	Alaska	65	1.82	-3.3	2.95	5.5	9.9	46.6	14.5	-1.07	0.15
FX-064	Site D	Sahara	24	2.85	-0.5	3.04	9.1	9.9	48.2	14.7	-0.95	0.63
FX-065	Site A	Northern Europe	58	1.11	-3.4	3.25	7.2	10.4	44.7	14.1	-1.14	-0.96
FX-066	Site A	Northern Europe	58	0.14	-12.5	3.91	5.7	11.2	49.1	14.5	-0.83	0.51
FX-067	Site D	Sahara	24	0.27	-7.9	3.32	6.1	10.2	47.2	13.6	-1.74	0.43
FX-068	Site F	Alaska	65	2.38	-6.0	3.56	6.4	10.5	45.8	13.6	-0.91	0.02
Fennec fox												
FX-106	Site B	Siberia	68	3.07	31.9	1.20	16.6	7.4	37.8	11.1	-0.54	-0.07
FX-107	Site A	Northern Europe	58	2.68	27.1	1.13	15.9	7.6	33.9	10.7	-0.64	-1.90
FX-108	Site D	Sahara	24	0.64	23.8	1.21	15.5	7.6	33.1	11.3	-0.49	-2.58
FX-109	Site B	Siberia	68	3.09	25.4	0.99	15.0	6.8	33.6	10.4	-1.50	-2.06
FX-110	Site D	Sahara	24	2.21	24.9	0.97	14.6	7.0	34.6	10.3	-1.07	-1.32
FX-111	Site E	Arabia	25	3.05	25.0	1.09	14.5	7.7	32.5	10.9	-1.16	-2.09
FX-112	Site B	Siberia	68	2.73	28.1	1.16	16.0	6.9	36.0	10.6	-1.16	-0.05
FX-113	Site D	Sahara	24	2.26	31.4	1.15	15.8	7.6	36.5	11.8	-0.20	-0.78

Skull shape scores are standardized (mean 0, SD 1) over all specimens.

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5 Discussion

The simulated data reproduce the qualitative pattern expected from the literature: larger bodies in the temperate species, and longer ears in warmer climates. The effect sizes carry no information because the data were generated from the same equations that were used to analyze them.

5.1 Limitations

- The data are fictional.
- Specimen age was drawn independently of site, so no spatial structure exists.
- The lineage process ignores extinction.

6 Conclusions

This document demonstrates numbered sections and subsections, cross-references to figures, tables, equations and appendices, a landscape page, separately numbered appendices, a bibliography and tables of contents, tables and figures.

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7 References

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A1 Appendix 1 - Model details

This appendix gives the coefficients of the two linear models of Section 3.2. Tables and figures are numbered per appendix.

A1.1 Body mass model

Table A1-1: Coefficients of the log body mass model (Equation 1). Reference species: red fox.

Term	Estimate	Std. error	t value	p value
Intercept (Red fox)	1.881	0.018	106.299	<0.0001
Age (Red fox)	-0.035	0.009	-3.819	0.0002
Arctic fox (offset)	-0.580	0.031	-18.544	<0.0001
Fennec fox (offset)	-1.687	0.035	-48.413	<0.0001
Age × Arctic fox	-0.017	0.016	-1.075	0.2840
Age × Fennec fox	-0.006	0.016	-0.360	0.7193

A1.2 Ear length model

Table A1-2: Coefficients of the ear length model on mean annual temperature. Reference species: red fox.

Term	Estimate	Std. error	t value	p value
Intercept (Red fox)	8.478	0.242	35.035	<0.0001
Temperature (Red fox)	0.134	0.022	6.013	<0.0001
Arctic fox (offset)	-0.652	0.291	-2.243	0.0264
Fennec fox (offset)	2.326	0.849	2.740	0.0069
Temperature × Arctic fox	0.037	0.037	0.998	0.3200
Temperature × Fennec fox	0.039	0.037	1.044	0.2983

A1.3 Skull shape

Skull shape scores separated the species along the first axis (Figure A1-1).

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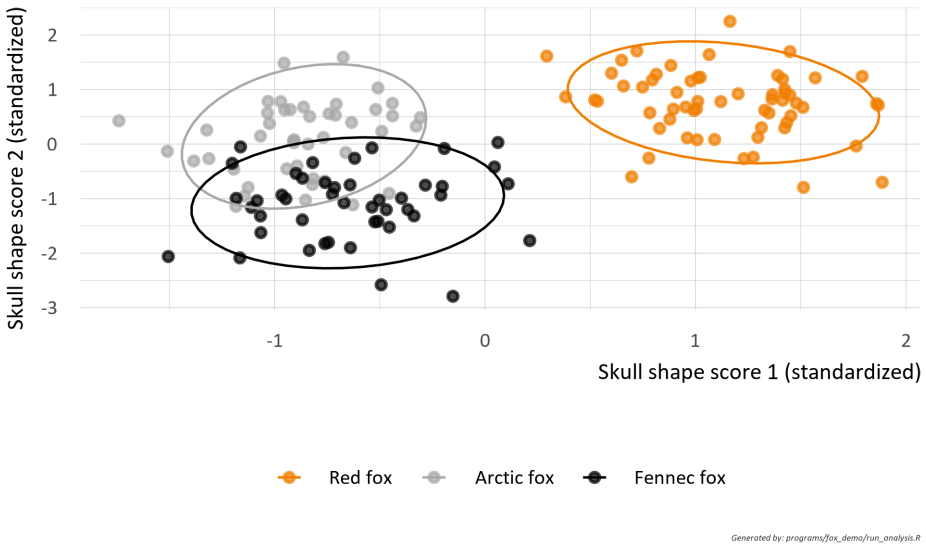


Figure A1-1: Skull shape scores 1 and 2 by species with 90% data ellipses.

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A2 Appendix 2 - Sampling and diagnostics

A2.1 Sampling sites

Specimens came from the sites listed in Table A2-1. Figure A2-1 shows a museum specimen of the Arctic fox.

Table A2-1: Number of simulated specimens by site and species.

Site	Region	Latitude (°N)	Red fox	Arctic fox	Fennec fox	Total
Site A	Northern Europe	58	9	10	8	27
Site B	Siberia	68	10	2	11	23
Site C	Greenland	72	8	6	6	20
Site D	Sahara	24	10	10	9	29
Site E	Arabia	25	18	10	8	36
Site F	Alaska	65	5	7	3	15



Figure A2-1: Arctic fox specimen at a natural history museum. Photograph by the World Museum vertebrate zoology curator, CC0, Wikimedia Commons.

A2.2 Model diagnostics

Figure A2-2 shows the residuals of the body mass model, and Table A2-2 the sample size and mean body mass per age class.

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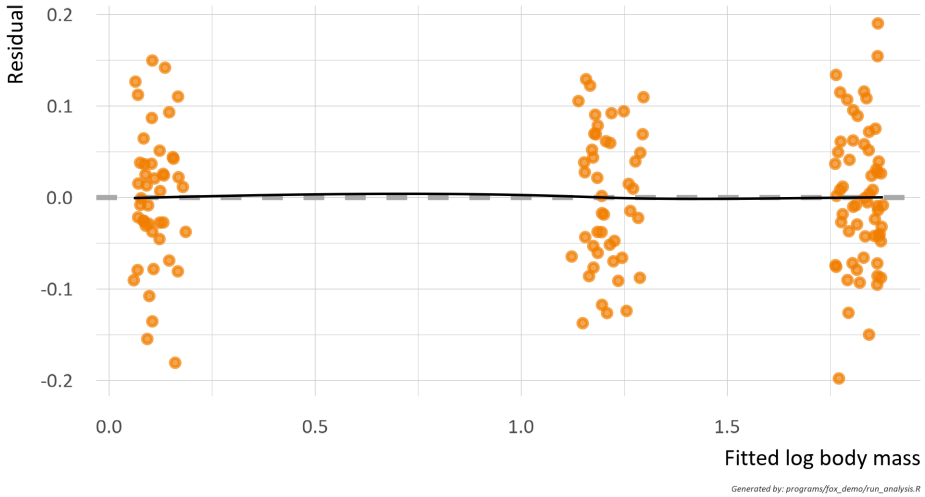


Figure A2-2: Residuals versus fitted values of the log body mass model.

Table A2-2: Number of specimens and mean body mass (kg) by age class and species.

	Red fox		Arctic fox		Fennec fox	
Age class (Ma)	**N**	**Mean mass**	**N**	**Mean mass**	**N**	**Mean mass**
0–1	22	6.46	10	3.62	8	1.18
1–2	15	6.28	11	3.35	10	1.16
2–3	12	6.02	22	3.26	21	1.09
3–3.5	11	5.90	2	3.17	6	1.08

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